

CREDIT OS (LENDR) — PRODUCT REQUIREMENTS DOCUMENT

1. PRODUCT OVERVIEW

Product Name:

Credit OS (Lendr)

Vision:

Empower merchants to offer **instant, embedded credit at checkout** through a plug-and-play API, increasing conversion rates while managing credit risk intelligently.

Problem Statement:

- Many Nigerian consumers cannot afford high-value purchases upfront.
- Merchants lose sales due to lack of financing options.
- Existing lending solutions are slow, fragmented, or require leaving checkout.

Solution:

A **real-time credit infrastructure layer** that:

- Verifies users instantly
 - Scores creditworthiness using alternative data
 - Offers flexible installment plans
 - Handles disbursement, repayment, and reporting
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2. GOALS & SUCCESS METRICS

Business Goals:

- Increase merchant conversion rate by 20–40%
- Maintain default rate < 8%
- Achieve < 5 seconds checkout decision time

Product Metrics:

- Loan approval rate
 - Default rate (30+ days overdue)
 - Repayment success rate
 - Time to decision (< 500ms for scoring)
 - Webhook delivery success rate
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3. USER PERSONAS

Merchant

“I want to embed a credit option so I can increase sales.”

Borrower

“I want to split payments easily without stress or leaving checkout.”

Lendr Ops

“I want to control risk, monitor loans, and ensure compliance.”

4. HIGH-LEVEL USER FLOW

1. Customer selects “Buy with Lendr”
2. Identity verification (BVN/NIN)
3. Bank data aggregation
4. Credit scoring

5. Offer generation
 6. User accepts + pays deposit
 7. Loan disbursed to merchant
 8. Repayment cycle begins
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5. CREDIT DECISION ENGINE

Structure:

- Identity Gate (Hard Reject)
- Capacity Score (40 pts)
- Credit History (25 pts)
- Behavioral Signals (20 pts)
- Stability (15 pts)

Decision Output:

- Score (0–100)
- Risk Tier (A–D)
- Loan Limit
- Terms

Tier Logic:

Tier	Score	Terms
A	75–100	0–10% downpayment
B	60–74	20% downpayment
C	45–59	40–50% downpayment
D	<45	Reject / micro-loan

HARD REJECTION LAYER (Before scoring)

If ANY of these hit → **AUTO REJECT**

From Dojah:

- BVN name mismatch
- Phone not linked to BVN
- DOB mismatch

From National Identity Management Commission:

- NIN verification fails (if required)

From CRC Credit Bureau / First Central Credit Bureau:

- Active default
- Blacklisted user

Fraud signals:

- 3 loan attempts in 24hrs
 - Device fingerprint reused across accounts
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CAPACITY SCORE (40 POINTS)

Using Okra / Mono

Metrics:

A. Income Strength (20 pts)

- Avg monthly inflow $\geq 4x$ installment → 20 pts
- $\geq 3x$ → 15 pts
- $\geq 2x$ → 10 pts
- $< 2x$ → 0 pts

B. Income Consistency (10 pts)

- Salary detected ≥ 3 months $\rightarrow 10$ pts
 - 2 months $\rightarrow 7$ pts
 - Irregular inflow $\rightarrow 3$ pts
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C. Expense Pressure (10 pts)

- Expenses $< 50\%$ income $\rightarrow 10$ pts
 - 50–70% $\rightarrow 5$ pts
 - 70% $\rightarrow 0$ pts
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CREDIT HISTORY SCORE (25 POINTS)

From CRC Credit Bureau / First Central Credit Bureau

A. Repayment History (15 pts)

- No missed payments $\rightarrow 15$ pts
 - 1–2 late payments $\rightarrow 8$ pts
 - Frequent late $\rightarrow 3$ pts
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B. Active Loans (10 pts)

- 0–1 loans $\rightarrow 10$ pts
 - 2–3 loans $\rightarrow 5$ pts
 - 3 loans $\rightarrow 0$ pts
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BEHAVIORAL SCORE (20 POINTS)

From bank + inferred patterns:

A. Airtime / Small Payments (10 pts)

- Frequent small payments → 10 pts
- Occasional → 5 pts
- None → 0 pts

👍 Proxy for **financial discipline**

B. Gambling Risk (5 pts)

- No betting transactions → 5 pts
 - <10% inflow → 2 pts
 - 30% → 0 pts
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C. Transaction Activity (5 pts)

- Daily/weekly activity → 5 pts
 - Sparse → 2 pts
 - Dormant → 0 pts
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STABILITY SCORE (15 POINTS)

A. Account Age (7 pts)

- 1 year → 7 pts
- 6–12 months → 4 pts
- <6 months → 0 pts

B. Employer Consistency (8 pts)

- Same employer ≥ 6 months $\rightarrow$ 8 pts
 - Changing frequently $\rightarrow$ 3 pts
 - Unknown $\rightarrow$ 0 pts
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FINAL SCORE CALCULATION

Total Score = Capacity + Credit + Behavioral + Stability
(Max = 100)

8. DECISION TIERS (ACTUAL OUTPUT)

Score	Tier	Decision
75–100	A	Approve (Best terms)
60–74	B	Approve (Moderate risk)
45–59	C	Approve (High downpayment)
<45	D	Reject

9. OFFER GENERATION LOGIC

Tier A

- 0–10% downpayment
 - 3–4 months tenor
 - Low interest
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Tier B

- 20% downpayment
 - 2–3 months
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Tier C

- 40–50% downpayment
 - 1–2 months
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Tier D

- Reject OR
 - Offer micro-loan (₦10k test loan)
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6. EDGE CASE HANDLING

- Missing bank data → fallback estimation
 - No credit history → rely on behavioral scoring
 - Multiple accounts → aggregate data
 - Payment failure → retry + state management
 - Merchant fraud → escrow + delayed settlement
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7. SYSTEM ARCHITECTURE

Core Components:

- API Gateway
- Identity/KYC Service
- Decision Engine
- Payment Service
- Core Database
- Worker Layer

- Notification Service

Design Principles:

- Modular monolith (Phase 1)
 - Async processing via events
 - Idempotent APIs
 - Fail-safe integrations
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8. DATABASE SCHEMA

Users

- id, name, email, phone
- bvn (encrypted), nin (encrypted)
- kyc_status, risk_flag

Loans

- id, user_id, merchant_id
- principal, interest, tenor
- status, risk_tier
- start_date, end_date

Repayments

- loan_id, installment_number
- due_date, amount_due, amount_paid
- status

Ledger (Source of Truth)

- transaction_type
- debit/credit
- amount
- reference_id
- immutable records

9. EVENT SYSTEM

Core Events:

- loan.created / approved / disbursed / defaulted
- payment.successful / failed
- repayment.due / paid / overdue

Architecture:

- Event producers (API, payments)
 - Message queue
 - Consumers:
 - Ledger
 - Loans
 - Notifications
 - Webhooks
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10. WEBHOOK SYSTEM

Purpose:

Notify merchants of real-time loan/payment updates

Events Sent:

- loan.approved
- loan.disbursed
- payment.failed
- loan.defaulted

Features:

- Retry logic (exponential backoff)

- Idempotency via event_id
 - Signature verification (HMAC)
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11. PAYMENT SYSTEM

Disbursement:

- Transfer to merchant account

Repayment:

- Card auto-debit
- Bank debit fallback

Failure Handling:

- Retry (3x)
 - Mark overdue
 - Trigger notifications
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12. RECONCILIATION SYSTEM

Daily Checks:

- Internal ledger vs payment provider
- Detect mismatches

Strategy:

- Immutable ledger
 - Reference-based matching
 - Automated alerts
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13. COMPLIANCE & REGULATION

Requirements:

- User consent for data access
- Secure storage of BVN/NIN
- Transparent loan terms (APR, fees)

Credit Reporting:

- Report all loans (good & bad repayment behavior)
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14. RISK & ANTI-DEFAULT STRATEGY

Before Disbursement:

- Strict scoring
- Tier-based limits

During Loan:

- Dynamic risk adjustment

Early Warning Signals:

- Payment delays
- Income changes
- Behavioral anomalies

Actions:

- Reminders
 - Restructuring offers
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15. API DESIGN (HIGH-LEVEL)

Core Endpoints:

- POST /v1/checkouts
 - POST /v1/verify
 - GET /v1/offers
 - POST /v1/loans
 - POST /v1/repayments
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16. NON-FUNCTIONAL REQUIREMENTS

Performance:

- < 500ms scoring
- < 5s total checkout

Reliability:

- 99.9% uptime target
- Retry mechanisms

Security:

- Encryption (PII)
 - Webhook signing
 - Access control
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17. KEY PRODUCT DECISIONS

- Rule-based scoring (Phase 1)
- Modular monolith architecture

- Paystack-first payment integration
 - Escrow model for merchants
 - Event-driven system for scalability
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18. FUTURE ROADMAP

Phase 2:

- ML-based fraud detection
- Dynamic credit limits
- Loan stacking detection

Phase 3:

- Cross-platform credit identity
 - BNPL marketplace
 - Embedded finance SDK
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FINAL NOTE

Credit OS is not just a lending product—it is a **financial infrastructure layer** designed for:

- Scalability
- Explainability
- Real-world Nigerian constraints
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